

Homework 4 (of 5)

Probability and Statistics

Please Note:

1. Clearly write your student name and ID.
2. No need to repeat writing the questions, just clearly label the question number.
3. Hard copy submission during the classes of **26 or 28 November**.
4. **Late submission is absolutely not accepted.**
5. Your assignment paper will not be returned. Please keep a copy for yourself.

5.16 A survey conducted by a song rating service finds that the percentages of listeners familiar with *Poker Face* by Lady Gaga who would give the song ratings of 5, 4, 3, 2, and 1 are, respectively, 43 percent, 21 percent, 22 percent, 7 percent, and 7 percent. Assign the numerical values 1, 2, 3, 4, and 5 to the (qualitative) ratings 1, 2, 3, 4, and 5 and find an estimate of the probability distribution of x this song's rating by a randomly selected listener who is familiar with the song.

Q2 In Exercise Q1,

- (a) Find the **expected value** of the estimated probability distribution.
- (b) Explain the meaning of this expected value in terms of all possible *Poker Face* listeners.

5.23

The customer service department for a wholesale electronics outlet claims that 90 percent of all customer complaints are resolved to the satisfaction of the customer. In order to test this claim, a random sample of 15 customers who have filed complaints is selected.

- a** Let x = the number of sampled customers whose complaints were resolved to the customer's satisfaction. Assuming the claim is true, write the binomial formula for this situation.
- b** Use the binomial tables (see Table A.1, page 783) to find each of the following if we assume that the claim is true:
 - (1) $P(x \leq 13)$.
 - (2) $P(x > 10)$.
 - (3) $P(x \geq 14)$.
 - (4) $P(9 \leq x \leq 12)$.
 - (5) $P(x \leq 9)$.
- c** Suppose that of the 15 customers selected, 9 have had their complaints resolved satisfactorily. Using part *b*, do you believe the claim of 90 percent satisfaction? Explain.

5.27

The January 1986 mission of the Space Shuttle *Challenger* was the 25th such shuttle mission. It was unsuccessful due to an explosion caused by an O-ring seal failure.

- a** According to NASA, the probability of such a failure in a single mission was $1/60,000$. Using this value of p and assuming all missions are independent, calculate the probability of no mission failures in 25 attempts. Then calculate the probability of at least one mission failure in 25 attempts.
- b** According to a study conducted for the Air Force, the probability of such a failure in a single mission was $1/35$. Recalculate the probability of no mission failures in 25 attempts and the probability of at least one mission failure in 25 attempts.
- c** Based on your answers to parts *a* and *b*, which value of p seems more likely to be true? Explain.
- d** How small must p be made in order to ensure that the probability of no mission failures in 25 attempts is .999?